

Introduction to Module 8: Graph Algorithms

Design and Analysis of Algorithms

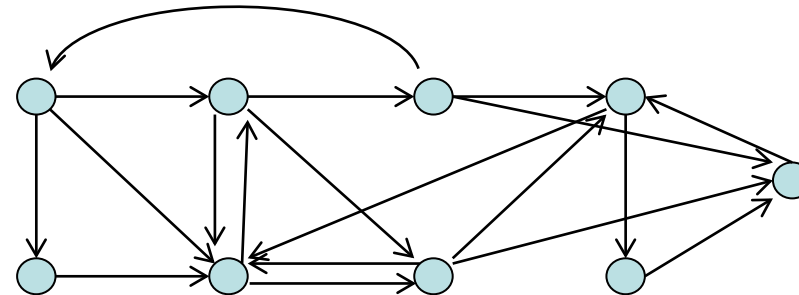
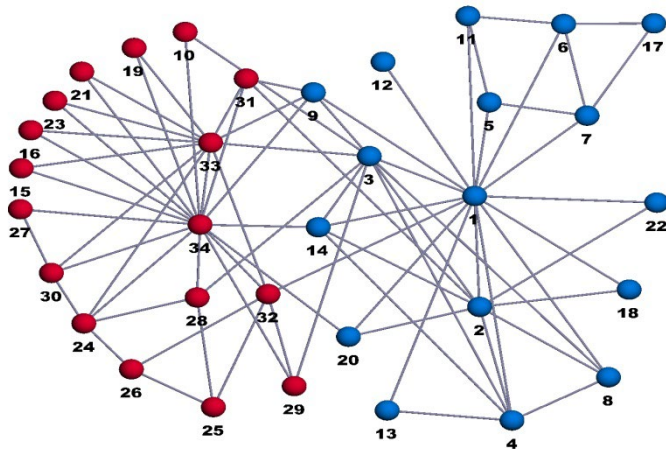
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Graphs (Networks)

- A **graph** is comprised of nodes (a.k.a. vertices) and edges. Each edge joins a pair of vertices.
- Graphs can be either undirected or directed; by default “graph” == “undirected graph”.
- Graphs are extremely common in CS and algorithms, since they can model a very wide range of problems and applications.



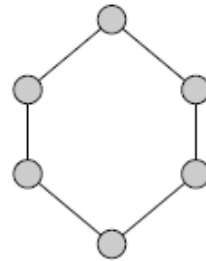
Graph Types

- Graph theorists have studied many different special types of graphs.
- Some simple examples of special classes of graphs we'll tend to see include:

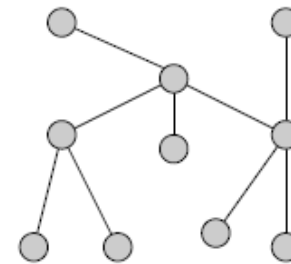
- The path (P_n)
- The cycle (C_n)
- The complete graph (K_n)
- Trees and forests
- Bipartite graphs
- Planar graphs



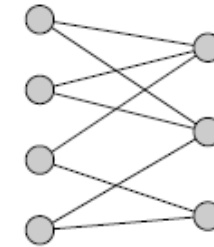
Path



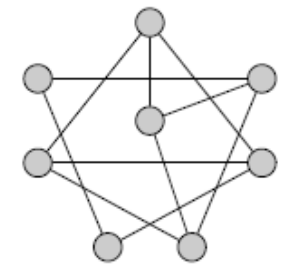
Cycle



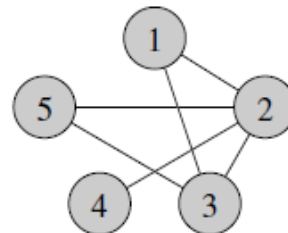
Tree



Bipartite Graph

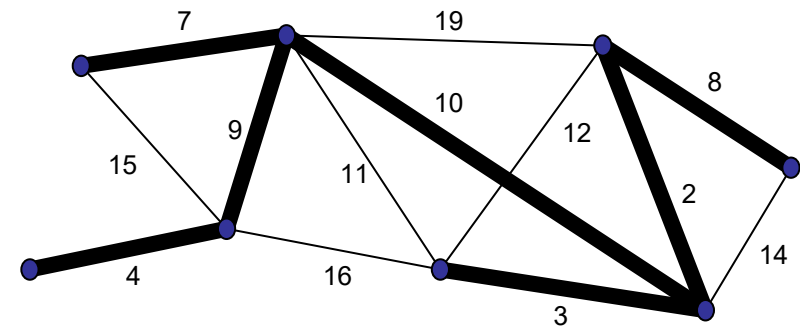
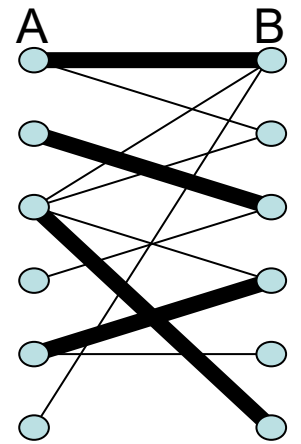


General Graph



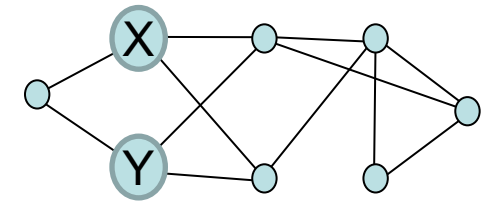
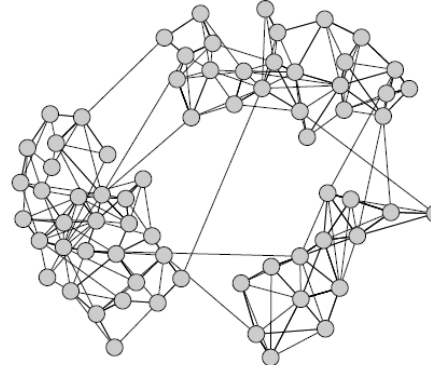
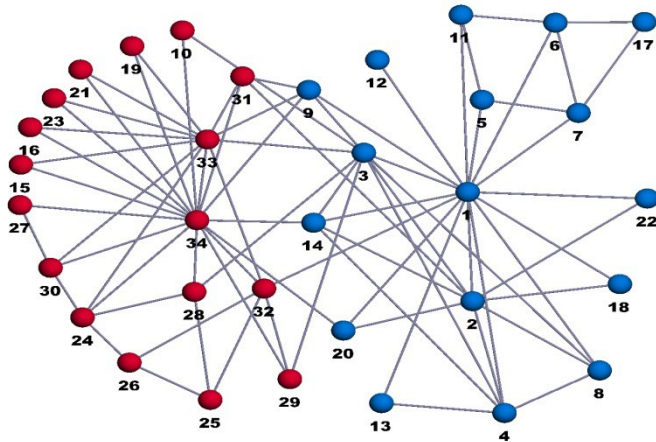
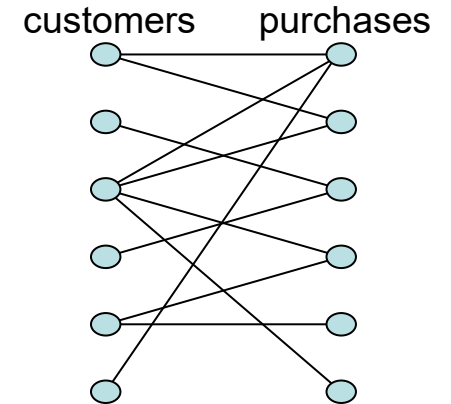
Some Classical Graph Problems

- [Shortest Paths]: Find the shortest path between two nodes, or from one node to all other nodes.
- [Minimum Spanning Trees]: Find a min-cost subset of edges that connects together all nodes.
- [Matchings]: Pair up as many nodes as possible, or pair up all nodes at minimum total cost.
- [Flow / Routing]: Route a maximum amount of some commodity through a capacitated network, possibly at minimum total cost.



Network Analysis / Data Mining

- [Similarity / Connectivity]: How similar are nodes X and Y if edges connect directly-related elements?
- [Clustering]: Does a graph break naturally into several large “clusters”?
- [Centrality]: Find nodes that are well-connected with all other nodes.

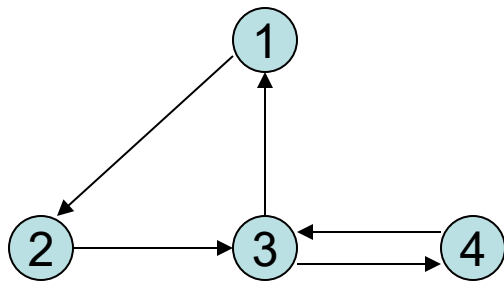


Terminology

- Walk: series of nodes connected by edges.
 - Also fine to think of it as a sequence of edges.
 - In a digraph, we tend to focus on directed walks.
- Path: a “simple” walk (visits no node twice).
- Cycle: a path that starts and ends at the same node.
- Connected components
- Strongly connected components in a digraph
 - Two nodes i and j belong to the same SCC if there's a directed path from i to j and from j to i .

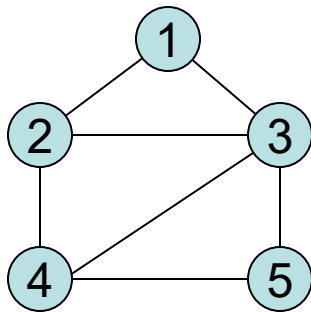
Graph Representation

- Two main ways to represent a graph:
 - An **adjacency matrix** (good for **dense** graphs)



	1	2	3	4
1	0	1	0	0
2	0	0	1	0
3	1	0	0	1
4	0	0	1	0

- **Adjacency lists** (ideal for **sparse** graphs)



1:	2	3		
2:	1	3	4	
3:	1	2	4	5
4:	2	3	5	
5:	3	4		

- Unless otherwise stated, we'll assume our graphs are represented using adjacency lists.

Main Topics

- Algorithms for connectivity, emphasizing “depth-first” recursive search.
- Shortest paths
- Coding practice implementing depth-first and breadth-first search
- Flows and matchings
- Enrichment discussion: “stable” matchings (highlighting more connections between algorithms and game theory).

Design and Analysis Assignment #2

- Similar in style to assignment #1
- May tend to emphasize new topic areas: discrete optimization and graphs